Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos\theta$ and $\tan\theta$.

Adjacent side: $\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\cos\theta = \frac{24}{25}$. $\tan\theta = \frac{7}{24}$.

2 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin\theta$ and the hypotenuse.

Hypotenuse: $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. $\sin\theta = \frac{15}{17}$.

3 **[Foundational]** • **Target: Exact values and the unit circle**

State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$. $\cos(45^\circ) = \frac{\sqrt{2}}{2}$.

4 **[Foundational]** • **Target: Exact values and the unit circle**

State the exact value of $\tan(60^\circ)$ and $\sin(30^\circ)$.

$\tan(60^\circ) = \sqrt{3}$. $\sin(30^\circ) = \frac{1}{2}$.

5 **[Foundational]** • **Target: Graphs of trigonometric functions**

State the amplitude and period of $y = 4\sin(2x)$.

Amplitude: 4. Period: $\frac{2\pi}{2} = \pi$.

6 **[Foundational]** • **Target: Graphs of trigonometric functions**

State the amplitude and period of $y = 3\cos\left(\frac{x}{2}\right)$.

Amplitude: 3. Period: $\frac{2\pi}{1/2} = 4\pi$.

7 **[Foundational]** • **Target: Trigonometric identities**

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Trigonometric identities

Simplify $\frac{\sin\theta}{1-\cos\theta} \times (1-\cos\theta)$.

The $(1-\cos\theta)$ factors cancel, leaving $\sin\theta$.

9 [Intermediate] • Target: Trigonometric identities

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin\theta = \frac{12}{13}$.

10 [Intermediate] • Target: Solving trigonometric equations

Solve $2\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

$\sin\theta = \frac{1}{2}$. Solutions: $\theta = 30^\circ$ or $\theta = 150^\circ$.

11 [Intermediate] • Target: Solving trigonometric equations

Solve $\cos(2\theta) = -1$ for $0^\circ \le \theta < 360^\circ$.

$2\theta = 180^\circ$ or $2\theta = 540^\circ$ (within $0^\circ \le 2\theta < 720^\circ$), so $\theta = 90^\circ$ or $\theta = 270^\circ$.

12 [Intermediate] • Target: Solving trigonometric equations

Solve $\tan\theta = -1$ for $0^\circ \le \theta < 360^\circ$.

$\theta = 135^\circ$ or $\theta = 315^\circ$.

13 [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle ABC, $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.

$c^2 = 9^2 + 12^2 - 2(9)(12)\cos(55^\circ) = 81 + 144 - 216(0.5736) \approx 225 - 123.9 \approx 101.1$. $c \approx \sqrt{101.1} \approx 10.1$.

14 [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle ABC, angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b, to the nearest tenth, using the Law of Sines.

$\frac{a}{\sin A} = \frac{b}{\sin B}$: $\frac{8}{\sin 40^\circ} = \frac{b}{\sin 70^\circ}$. $b = \frac{8\sin 70^\circ}{\sin 40^\circ} \approx \frac{8(0.9397)}{0.6428} \approx 11.7$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 19.01 Normal Distribution

In triangle ABC, $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.

The largest angle is opposite the longest side, so we find angle C:

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 100 - 169}{2(7)(10)} = \frac{-20}{140} \approx -0.1429$. $C = \cos^{-1}(-0.1429) \approx 98^\circ$.

16 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.

$\sin(75^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

17 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 3.01 Graphs & Transformations

Use the identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$ to find the exact value of $\cos(15^\circ)$, using $15^\circ = 45^\circ - 30^\circ$.

$\cos(15^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

18 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 4.01 Coordinate Geometry

This question has two parts. A ship observes a lighthouse at an angle of elevation of 18°. After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30°. (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h, to the nearest meter.

(a) At the closer position: $\tan(30^\circ) = \frac{h}{d}$, so $h = d\tan(30^\circ)$. At the farther position (distance $d + 500$): $\tan(18^\circ) = \frac{h}{d + 500}$, so $h = (d + 500)\tan(18^\circ)$. (b) Setting equal: $d\tan(30^\circ) = (d + 500)\tan(18^\circ)$. Using $\tan(30^\circ) \approx 0.5774$ and $\tan(18^\circ) \approx 0.3249$: $0.5774d = 0.3249d + 162.45$, so $0.2525d = 162.45$, giving $d \approx 643.4$. Then $h = 643.4 \times 0.5774 \approx 372$ meters.

19 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 6.01 Differentiation & Integration

Describe fully the single transformation that maps $y = x^2$ to $y = (\frac{x}{3})^2$.

Final Answer: Stretch parallel to the x-axis, scale factor 3. Comparing the two equations term-by-term identifies the transformation as: Stretch parallel to the x-axis, scale factor 3.

20 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 7.01 Statistics & Probability

Describe fully the single transformation that maps $y = x^2$ to $y = (\frac{x}{4})^2$.

Final Answer: Stretch parallel to the x-axis, scale factor 4. Comparing the two equations term-by-term identifies the transformation as: Stretch parallel to the x-axis, scale factor 4.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exact Trig Ratios & Reference Angles

Find the exact value of $\sin(210^\circ)$.

a. $-\frac{1}{2}$ ✓ **b.** $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$ **d.** $\frac{\sqrt{3}}{2}$

210° is in Quadrant III with reference angle 30°. Sine is negative in QIII, so $\sin(210^\circ) = -\sin(30^\circ) = -1/2$.

22 [Foundational] • Target: Exact Trig Ratios & Reference Angles

Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. $-\frac{\sqrt{2}}{2}$ ✓ **b.** $\frac{\sqrt{2}}{2}$

c. $-\frac{1}{2}$ **d.** $\frac{\sqrt{3}}{2}$

$3\pi/4$ is in QII where cosine is negative: $\cos(3\pi/4) = -\cos(\pi/4) = -\sqrt{2}/2$.

23 [Foundational] • Target: Exact Trig Ratios & Reference Angles

If θ is reflex and $\tan(\theta) = \frac{4}{3}$, find the exact value of $\sin(\theta)$.

a. $-\frac{4}{5}$ ✓ **b.** $\frac{4}{5}$

c. $-\frac{3}{5}$ **d.** $\frac{3}{5}$

Since $\tan(\theta) > 0$ and θ is reflex, θ lies in Quadrant III. Here $x = -3, y = -4, r = 5$. Thus $\sin(\theta) = -4/5$.

24 [Foundational] • Target: Fundamental Trigonometric Identities

Simplify the expression $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all valid θ.

a. **1** ✓ **b.** $\sin^2\theta$

c. $\tan^2\theta$ **d.** $\sec^2\theta$

$1 - \cos^2\theta = \sin^2\theta$. $1 + \cot^2\theta = \csc^2\theta = 1/\sin^2\theta$. Product is $\sin^2\theta \times (1/\sin^2\theta) = 1$.

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25 [Foundational] • Target: Fundamental Trigonometric Identities

Prove the identity $\frac{\sin\theta}{1-\cos\theta} + \frac{1-\cos\theta}{\sin\theta} \equiv 2\csc\theta$. What is the common numerator after combining fractions?

a. $2(1-\cos\theta)$ ✓

b. $\sin^2\theta + \cos^2\theta$

c. $2\sin\theta$

d. $1 - 2\cos\theta$

$\frac{\sin^2\theta + (1-\cos\theta)^2}{\sin\theta(1-\cos\theta)} = \frac{\sin^2\theta + 1 - 2\cos\theta + \cos^2\theta}{\sin\theta(1-\cos\theta)} = \frac{2-2\cos\theta}{\sin\theta(1-\cos\theta)} = \frac{2(1-\cos\theta)}{\sin\theta(1-\cos\theta)} = 2\csc\theta.$

26 [Foundational] • Target: Fundamental Trigonometric Identities

Given $\cos\theta = -\frac{5}{13}$ and $90^\circ < \theta < 180^\circ$, find $\tan\theta$.

a. $-\frac{12}{5}$ ✓

b. $\frac{12}{5}$

c. $-\frac{5}{12}$

d. $\frac{13}{12}$

In QII, sine is positive: $\sin\theta = \sqrt{1 - 25/169} = 12/13$. Then $\tan\theta = \frac{12/13}{-5/13} = -12/5$.

27 [Foundational] • Target: Solving Trigonometric Equations

Solve $2\sin(2x) = 1$ for $0^\circ \le x \le 180^\circ$.

a. $x = 15^\circ$ **and** $x = 75^\circ$ ✓

b. $x = 30^\circ$ and $x = 150^\circ$

c. $x = 15^\circ$ and $x = 105^\circ$

d. $x = 45^\circ$ and $x = 135^\circ$

$\sin(2x) = 1/2$. For $0^\circ \le 2x \le 360^\circ$: $2x = 30^\circ$ or $150^\circ \Rightarrow x = 15^\circ$ or 75°.

28 [Foundational] • Target: Solving Trigonometric Equations

Solve $2\cos^2\theta - 5\cos\theta + 2 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ$ **and** $\theta = 300^\circ$ ✓

b. $\theta = 30^\circ$ and $\theta = 330^\circ$

c. $\theta = 60^\circ$ and $\theta = 120^\circ$

d. $\theta = 45^\circ$ and $\theta = 315^\circ$

Factor: $(2\cos\theta - 1)(\cos\theta - 2) = 0$. Since $\cos\theta \le 1$, $\cos\theta = 1/2 \Rightarrow \theta = 60^\circ$ or $360^\circ - 60^\circ = 300^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Solving Trigonometric Equations

Solve $3\tan(x) = 2\sin(x)$ for $0 < x < 2\pi$.

a. $x = \pi$ **(excluded) or** $\cos x = 3/2$ **(no solution)** ✓

b. No solutions in $(0, 2\pi)$

c. $x = \pi/3$ and $5\pi/3$

d. $x = \pi/6$ and $5\pi/6$

$3\frac{\sin x}{\cos x} - 2\sin x = 0 \Rightarrow \sin x(3 - 2\cos x) = 0$. $\sin x = 0$ gives $x = \pi$ (since domain is open interval $(0, 2\pi)$, $x = \pi$ is a root!). $2\cos x = 3 \Rightarrow \cos x = 1.5$ impossible. Root is $x = \pi$.

30 [Intermediate] • Target: Trigonometric Graphs & Transformations

State the amplitude and period of the function $y = 3\sin(2x) - 1$.

a. **Amplitude** $= 3$**, Period** $= \pi$ ✓

b. Amplitude $= 3$, Period $= 2\pi$

c. Amplitude $= 2$, Period $= \pi$

d. Amplitude $= 4$, Period $= \pi$

Amplitude is $|a| = 3$. Period is $2\pi/b = 2\pi/2 = \pi$.

31 [Intermediate] • Target: Trigonometric Graphs & Transformations

Find the range of the function $f(x) = 4 - 3\cos(x)$ for all real x.

a. $[1, 7]$ ✓

b. $[-1, 7]$

c. $[1, 4]$

d. $[-3, 3]$

Since $-1 \leq \cos(x) \leq 1$: minimum is $4 - 3(1) = 1$, maximum is $4 - 3(-1) = 7$. Range is $[1, 7]$.

32 [Intermediate] • Target: Trigonometric Graphs & Transformations

Find the number of solutions to the equation $\sin(2x) = \cos(x)$ in the interval $0 \leq x \leq 2\pi$.

a. **4 solutions** ✓

b. 2 solutions

c. 3 solutions

d. 6 solutions

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \pi/2, 3\pi/2$. $\sin x = 1/2 \Rightarrow x = \pi/6, 5\pi/6$. Total 4 solutions.

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33 [Intermediate] • Target: Right-triangle trigonometry

Describe fully the single transformation that maps $y = x^2$ to $y = \left(\frac{x}{2}\right)^2$.

a. Stretch parallel to the x-axis, scale factor $\frac{1}{2}$

b. **Stretch parallel to the x-axis, scale factor** 2 ✓

c. Stretch parallel to the y-axis, scale factor 2

d. Translation parallel to the x-axis

Comparing the two equations term-by-term identifies the transformation as: Stretch parallel to the x-axis, scale factor 2.

34 [Intermediate] • Target: Exact values and the unit circle

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.

a. (3,6)

b. (6,-3)

c. **(-3,6)** ✓

d. (-3,-6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3, 6)$.

35 [Intermediate] • Target: Graphs of trigonometric functions

The minimum point of $y = x^2$ is translated by the vector $(-6; 5)$ to give curve C. State the coordinates of the minimum point of C.

a. (-6,-5)

b. (5,-6)

c. **(-6,5)** ✓

d. (6,5)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6, 5)$.

36 [Intermediate] • Target: Trigonometric identities

The minimum point of $y = x^2$ is translated by the vector $(1; -1)$ to give curve C. State the coordinates of the minimum point of C.

a. (-1,1)

b. (1,1)

c. **(1,-1)** ✓

d. (-1,-1)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(1, -1)$.

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37 [Intermediate] • Target: Solving trigonometric equations

The minimum point of $y = x^2$ is translated by the vector $(6;\ -5)$ to give curve C. State the coordinates of the minimum point of C.

a. (6,-5) ✓

b. (-6,-5)

c. (6,5)

d. (-5,6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(6,\ -5)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 4.01 Coordinate Geometry

The minimum point of $y = x^2$ is translated by the vector $(-2; 3)$ to give curve C. State the coordinates of the minimum point of C.

a. (-2,3) ✓

b. (-2,-3)

c. (3,-2)

d. (2,3)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-2, 3)$.

39 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 6.01 Differentiation & Integration

The minimum point of $y = x^2$ is translated by the vector $(1;\ -7)$ to give curve C. State the coordinates of the minimum point of C.

a. (-1,-7)

b. (1,-7) ✓

c. (-7,1)

d. (1,7)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(1,\ -7)$.

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40 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 7.01 Statistics & Probability

The minimum point of $y = x^2$ is translated by the vector $(-3; 1)$ to give curve C. State the coordinates of the minimum point of C.

a. (1,-3)

b. (-3,-1)

c. **(-3,1)** ✓

d. (3,1)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3, 1)$.

41 [Challenging] • Target: Exact Trig Ratios & Reference Angles • Cross-Lesson: 8.01 Functions: Composite & Inverse

The minimum point of $y = x^2$ is translated by the vector $(4; -4)$ to give curve C. State the coordinates of the minimum point of C.

a. (-4,4)

b. (4,4)

c. **(4,-4)** ✓

d. (-4,-4)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(4, -4)$.

42 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 9.01 Circular Measure

The minimum point of $y = x^2$ is translated by the vector $(7; -6)$ to give curve C. State the coordinates of the minimum point of C.

a. (7,6)

b. (-6,7)

c. (-7,-6)

d. **(7,-6)** ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(7, -6)$.

43 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(4; -7)$. State the coordinates of the new vertex.

a. (-4,-7)

b. **(4,-7)** ✓

c. (-7,4)

d. (4,7)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(4, -7)$.

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44 [Challenging] • Target: Trigonometric Graphs & Transformations • Cross-Lesson: 10.01 Series & Binomial Theorem

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(1;\ -6)$. State the coordinates of the new vertex.

a. (-1,-6)

b. **(1,-6) ✓**

c. (1,6)

d. (-6,1)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(1,\ -6)$.

45 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 11.01 Kinematics

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(7;\ -4)$. State the coordinates of the new vertex.

a. **(7,-4) ✓**

b. (7,4)

c. (-7,-4)

d. (-4,7)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(7,\ -4)$.

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